

Ian Leung

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EDUCATION

University of Waterloo

Expected 2028

Bachelor of Computer Science and Finance - Double Major

- President's Scholarship of Distinction (\$5000), Undergraduate Achievement Award (\$1500)
- Statistics Minor

Relevant Coursework: Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Calculus, Probability, Proofs, Financial Markets and Data Analytics

TECHNICAL SKILLS

Languages Python, Java, C++, C#, C, Go, Kotlin, TypeScript, SQL, HTML, CSS, Bash
Technologies Temporal, PyTorch, Supabase, Firebase, MongoDB, React, Maven, Qt, OpenRouter, OpenCV
Developer Tools Linux, Git, Docker, Jupyter, DigitalOcean, Modal, Render, Vercel, Unity, Android Studio

WORK EXPERIENCE

Roblox

Jun 2026 – Present

Software Engineer Intern | Go, Python

San Mateo, CA

- Building an automated CI/CD pipeline that deploys configuration changes across **5,000+** network devices in global datacenters using **GitHub Actions** and **Temporal**, replacing a slow, manual, push process
- Applying network programming to design a **canary** rollout workflow with automated health checks (**VictoriaMetrics**) and staged expansion to validate changes before fleet-wide deployment to Roblox's **100M+** daily active users
- Implementing deployment safety controls including **leased locks**, automatic **rollback**, and error classification, so parallel workflows are safe and devices self-recover to a known-good state on failure

Sunnybrook Research Institute

Jul 2025 – Aug 2025

Software Research Intern | Python

Toronto, ON

- Built an event tracing and playback tool to simulate medical professionals' interactions with **Focused Ultrasound** 🔄 software, enabling realistic testing and diagnosis of software defects, unintended user behaviours, and edge cases
- Created an automated software testing framework with **PyQt** and **Pydantic**, generating synthetic and replayable JSON event flows, reducing software testing times by **91%**
- Implemented a logging API using **Difflib** and **Loguru** to validate test results against baselines, identifying **5+** bugs

PROJECTS

🌐 **Data Science Club Website** | TypeScript, React, Supabase, HTML, CSS

- Maintained www.uwdatascience.ca, UWaterloo's data science club website serving **600+** members
- Replaced the executive team's manual term-start **Google Form** with an in-app onboarding flow backed by a normalized **Supabase/Postgres** schema that autofills profile and role data from existing records

📱 **Pharmagotchi: Pharmacy Companion App** | Android Studio, Figma, Kotlin, XML, OpenRouter

- Built a native **Android** (Kotlin) health-companion app where a virtual pet responds to missed medication doses, concerning vitals, and skipped logging, backed by an offline **Room** database and **WorkManager** background reminders
- Integrated **LLMs via OpenRouter** to power a context-aware chat assistant and auto-build each user's tracked metrics, parsing their conditions and medications into structured **JSON** vital signs

📱 **PeerAssist - Full-Stack Peer Reviewing** | Java, MongoDB, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for 20+ students to upload, review, and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with **Apache PDFBox** and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin

📱 **Trust No Ghost - 3D Maze Exploration Game** | Unity, C#, Itch.io

- Shipped Trust No Ghost 🎮 for **60+** users at the UW Game Jam Fall 2025, placing in the **top 10** most rated games
- Harnessed Unity and C# to develop expanding maze designs, AI pathfinding algorithms, and dynamic moving mechanics